Alaa Ali Mousa

Flutter Developer | Cross-Platform Mobile Engineer | AI-Integrated Applications

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Summary

Flutter Developer with hands-on experience building cross-platform mobile applications using **Dart**, **Flutter**, and **Firebase**, plus full-stack experience integrating **FastAPI** backends and **AI/deep-learning models** into production-style mobile systems. Led the mobile and system-integration work on *IntraVision*, a full-stack AI medical imaging application, from architecture through deployment. Skilled in **Clean Architecture**, state management (**Provider**, **Riverpod**, **BLoC**), **RESTful API** integration, and real-time data, with a strong focus on performance, UX, and end-to-end system design.

Education

Delta Technology University — B.Sc. in Information Technology and Software *Sept 2022 – June 2026*
Overall Grade: Very Good | Achieved Distinction (Excellence Appreciation) in three academic years

Skills

Languages & Frameworks: Flutter, Dart, Python, FastAPI, SQL, JSON
Mobile Architecture: Clean Architecture, BLoC/Cubit, Provider, Riverpod, setState, Dependency Injection (get_it)
Backend & Data: Firebase (Auth, Firestore, FCM), MongoDB, REST API Design, JWT Authentication, OOP
AI & Integration: AI/Deep Learning Model Integration, Third-Party API Integration, Data Processing
UI/UX: Component-Based UI, Responsive Design, Light/Dark Mode, Localization (Arabic/English)
Tools & Practices: Git, GitHub, Postman, Android Studio, Visual Studio Code, Agile Development

Experience

Flutter Developer Trainee — Digital Egypt Pioneers Initiative (DEPI) *2025 – 2026*
Ministry of Communications and Information Technology (MCIT), Egypt

- Completed an intensive, project-based Flutter Developer training program covering cross-platform mobile development and modern software engineering practices.
- Built and shipped multiple Flutter applications applying Firebase (Authentication, Firestore), RESTful API integration, and state management with Provider and Riverpod.
- Collaborated in team-based projects, strengthening communication, problem-solving, and time-management skills in an Agile workflow.

Projects

IntraVision — AI-Powered Brain Tumor Segmentation System *Graduation Project, 2025 – 2026*
Full-Stack Developer — Flutter, FastAPI, AI Integration

- Built a full-stack medical imaging platform enabling doctors to upload MRI scans, auto-generate contrast-enhanced images via deep learning, segment brain tumors, and produce AI-written medical reports from a mobile app.
- Developed the Flutter Android app with Clean Architecture and BLoC/Cubit state management, including biometric login, secure storage, and push notifications (Firebase Cloud Messaging).
- Integrated a CycleGAN model to synthesize contrast-enhanced (T1CE) MRI from native T1 scans and an nnU-Net v2 pipeline for automated tumor segmentation, returning confidence score, volume, and diameter metrics.
- Built a FastAPI backend with MongoDB Atlas, JWT authentication, 2FA email verification, and 32+ REST endpoints; generated AI-written PDF reports using Google Gemini and WeasyPrint.
- Implemented interactive 3D brain/tumor visualization (trimesh, model_viewer_plus) and a QR-code-based patient portal for real-time case tracking.
- Diagnosed and resolved critical production issues, including native crashes on large NIfTI file handling and out-of-memory errors on 150MB+ downloads, by redesigning the file I/O pipeline.

Flower App (Flutter)

- Built an e-commerce app with cart, favourites, and product filtering, featuring smooth animations and a responsive UI.

News App (Flutter)

- Built a news app fetching live global news via a REST API, with smooth navigation and caching for better performance.

Chat App (Flutter + Firebase)

- Built a real-time chat app using Firebase Authentication and Firestore, with message timestamps, chat rooms, and user login.

Weather App (Flutter)

- Built a weather app using the OpenWeatherMap API with a dynamic UI that adapts to temperature and conditions.

Smart Irrigation System (IoT)

- Built an automated irrigation system with Arduino, water pumps, and moisture sensors for real-time monitoring and control.

Courses & Certifications

- **Flutter & Dart** — Cerativa Menof (2025): Cross-platform mobile development and state management fundamentals.
- **Flutter & Dart** — Easy Learn Academy (2025): Real-world app development with Firebase integration and REST APIs.
- **Artificial Intelligence** — Huawei ICT Academy (2025): Machine learning fundamentals and AI workflow applications.
- **Data Science** — Cisco (2024): Data analysis, processing, and statistical fundamentals.
- **Internet of Things (IoT)** — Cisco (2024): Sensors, controllers, and connected systems for automation.

Languages

Arabic (Native) | English (Intermediate)